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4)

For any fixed state s we define $x = \pi^*(s)$, $y = \pi'(y)$

Hence $\Rightarrow Q_{V_{opt}}(s,y) \geq Q_{V_{opt}}(s,x)$

from Bellman's we expand both sides:

$$R(s,y) + \gamma \sum_{s'} P(s' \mid s,a) V_{opt}(s') \geq R(s,x) + \gamma \sum_{s'} P(s' \mid s,x) V_{opt}(s')$$

$$\Rightarrow R(s,y) - R(s,x) \geq \gamma \left(\sum_{s'} P(s' \mid s,a) V_{opt}(s') - \sum_{s'} P(s' \mid s,a) V_{opt}(s') \right)$$

By Hint: $V_{opt}(s') \leq V^*(s') + \varepsilon$ → epsilon in the first sigma on the right and

$V_{opt}(s') \geq V^*(s') - \varepsilon$ for second sigma:

$$R(s,x) - R(s,y) \leq \gamma \left(\sum_{s'} P(s' \mid s,y) V^*(s') + \varepsilon - \left(\sum_{s'} P(s' \mid s,x) V^*(s') - \varepsilon \right) \right)$$

$$\Rightarrow \boxed{R(s,x) - R(s,y) \leq \gamma \left(\sum_{s'} \left(P(s' \mid s,y) - P(s' \mid s,x) \right) V^*(s') + 2\varepsilon \right)}$$ result 1

now let k be an state ($k \in S$) with worst Loss:

$$L_{\pi'}(k) = \max_{s \in S} L_{\pi'}(s) \Longrightarrow L_{\pi'}(s') \leq L_{\pi'}(k)$$

and we know

$$V^*(k) = R(k,x) + \gamma \sum_{s'} \varphi(s' \mid k,x)\, V^*(s')$$

$$V^{\pi'}(k) = R(k,y) + \gamma \sum_{s'} \varphi(s' \mid k,y)\, V^{\pi'}(s')$$

$$L_{\pi'}(k) = V^*(k) - V^{\pi'}(k) = R(k,x) - R(k,y) + \gamma \left(\sum_{s'} \varphi(s' \mid k,x) V^*(s') - \sum_{s'} \varphi(s' \mid k,y) V^{\pi'}(s') \right)$$

with adding and subtracting $\sum_{s'} \varphi(s' \mid k,y) V^* s'$ we have:

$$L_{\pi'}(k) = R(k,x) - R(k,y) + \gamma \left(\sum_{s'} \left(\varphi(s' \mid k,x) - \varphi(s' \mid k,y) \right) V^*(s') \right) + \gamma \sum_{s'} \varphi(s' \mid k,y) \left(V^*(s') - V^{\pi'}(s') \right)$$

$$= R(k,x) - R(k,y) + \gamma \left(\sum_{s'} \left(\varphi(s' \mid k,x) - \varphi(s' \mid k,y) \right) V^*(s') \right) + \gamma \sum_{s'} \varphi(s' \mid k,y)\, L_{\pi'}(s')$$

From Result 1:

$$L_{\pi'}(k) \leq \gamma \left(\sum_{s'} \left(\varphi(s' \mid k,y) - \varphi(s' \mid k,x) \right) V^*(s') + 2\varepsilon \right) + \gamma \left(\sum_{s'} \left(\varphi(s' \mid k,x) - \varphi(s' \mid k,y) \right) V^*(s') \right)$$

$$+ \gamma \sum_{s'} \varphi(s' \mid k,y)\, L_{\pi'}(s') = 2\gamma\varepsilon + \gamma \sum_{s'} \varphi(s' \mid k,y)\, L_{\pi'}(s')$$

and since $L_{\pi'}(s') \leq L_{\pi'}(k)$ for all s':

$$\sum_{s'} \varphi(s' \mid k,y)\, L_{\pi'}(s') \leq \sum_{s'} \varphi(s' \mid k,y)\, L_{\pi'}(k) = L_{\pi'}(k) \rightarrow \boxed{\text{sum of probability} = 1}$$

$$\Longrightarrow L_{\pi'}(k) \leq 2\gamma\varepsilon + \gamma L_{\pi'}(k) \Longrightarrow L_{\pi'}(k) \leq \frac{2\gamma\varepsilon}{1-\gamma}$$

since $L_{\pi'}(k) = \max_s L_{\pi'}(s)$

$$\Longrightarrow \max_s L_{\pi'}(s) \leq \frac{2\gamma\varepsilon}{1-\gamma}$$

which shows a good approximation could result in a good policy with a bounded loss on every state.